

Reviewer Pack — Minimal Order of K-theory and Communication Complexity Rank ...

Entry b216b6b59fbf · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of K-theory and Communication Complexity Rank Variance

Entry ID: b216b6b59fbf **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-06 03:25:20 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (K-theory) **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For any given n -vertex graph G , the minimal order of the finite abelian group associated with its K-theory ($\min_order(K(G))$) is linearly correlated with its communication complexity rank variance ($rank_var(G)$), such that $\min_order(K(G)) = \Theta(rank_var(G))$.

Rationale (proposer's reasoning):

K-theory provides a topological invariant that captures information about vector bundles over a space, which could potentially encode structural properties related to the flow of information in communication networks. The communication complexity rank variance measures the diversity of communication patterns among different pairs of vertices in a graph. This conjecture suggests that the complexity of encoding these communication patterns may be linked to the algebraic structure provided by K-theory.

Taxonomy category: K-theory_and_Communication_Complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 3096fb13e2c3aee6

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if at least 80% of the Jaccard similarity coefficients between $\min_order(K(G))$ and $rank_var(G)$ are above 0.5, across all 30 seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "K-theory" AND "communication complexity" AND "matrix rank" - "finite abelian group" AND "graph K-theory" AND "communication complexity" - "algebraic geometry" AND "minimal order K-theory" AND "rank variance communication complexity"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.2s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_graph(n):
        edges = set()
        for i in range(n):
            for j in range(i + 1, n):
                if random.random() < 0.5:
                    edges.add((i, j))
        return edges

    def k_theory(edges):
        # Simplified K-theory calculation (not actual K-theory)
        return len(edges) % 2

    def communication_complexity_rank_variance(graph):
        n = len(graph)
        rank_var = 0
        for i in range(n):
            for j in range(i + 1, n):
                if (i, j) in graph:
                    rank_var += 1
        return rank_var / (n * (n - 1) / 2)

    def min_order(K):
        # Simplified minimal order calculation
        return K

    n_values = [5, 10, 15, 20, 30, 40]
```

```

results = []

for n in n_values:
    graph = generate_graph(n)
    K = k_theory(graph)
    min_K = min_order(K)
    rank_var = communication_complexity_rank_variance(graph)
    results.append((min_K, rank_var))

if not results:
    return {
        "metric_name": "Jaccard_similarity",
        "metric_value": 0.0,
        "instances_tested": 0,
        "n_max": 0,
        "conjecture_holds": False,
        "counterexample": "No instances generated"
    }

min_Ks, rank_vars = zip(*results)
jaccard_similarity = sum(abs(x - y) for x, y in zip(min_Ks, rank_vars)) / len(results)

return {
    "metric_name": "Jaccard_similarity",
    "metric_value": jaccard_similarity,
    "instances_tested": len(results),
    "n_max": max(n_values),
    "conjecture_holds": jaccard_similarity > 0.5,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    if not results:
        print("RESULT: INCONCLUSIVE no trials executed")
        sys.exit(0)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(s for s, r in zip(seeds, results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"Jaccard similarity below 0.5\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE insufficient support")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
27321, 'instances_tested': 6, 'n_max': 40, 'conjecture_holds': False, 'counterexample': ''}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.5444894454114312, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.5932667260230923, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.3867035953395767, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.5901175352173538, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.4868984469472975, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.4279854020898323, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.02832562768044829, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.673618593927729, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.409010741585817, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.40811830750974415, 'instances_tested': 6}
TRIAL: {'metric_name': 'Jaccard_similarity', 'metric_value': 0.4173943897264114, 'instances_tested': 6}
RESULT: FALSIFIED counterexample="Jaccard similarity below 0.5" first_failing_seed=23
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test uses a simplified version of K-theory and communication complexity rank variance, which may not accurately represent the actual mathematical concepts. The metric saturation is likely due to the trivial calculation of K-theory ($\text{len}(\text{edges}) \% 2$), which does not scale with n and will always result in 0 or 1, making it impossible for $\text{min_order}(K)$ to be linearly correlated with $\text{rank_var}(G)$.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The Jaccard similarity coefficient fraction was below 0.8, indicating that less than 80% of the coefficients were above 0.5, which is the falsificatio | next: Further investigation is needed to understand why the simplified version of K-theory does not correlate with rank variance as expected. Consider using a more accurate representation of K-theory and communication complexity rank variance in future tests.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14085
2	preregistration	ollama_remote	glm4:latest	0	0	9482
3	novelty	ollama_remote	glm4:latest	0	0	9444
4	novelty	ollama_remote	glm4:latest	0	0	8815
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	24972
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14938
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16496

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10477
9	critic	ollama_remote	glm4:latest	0	0	15786
10	judge	ollama_remote	glm4:latest	0	0	9393

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 133888 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/b216b6b59fbf.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/b216b6b59fbf.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/b216b6b59fbf.tar.gz (if generated)